A Forgotten Particle May Explain Why the Universe Exists

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Scientists have a new idea about why the universe exists at all. Long ago, matter and **antimatter** were made in almost equal amounts. When they meet, they destroy each other. If the amounts had been exactly equal, almost nothing would be left. There would be no stars, planets, or life.

Researchers from the University of Southampton and the Indian Institute of Technology Kanpur say a forgotten particle may have helped. It is a heavier cousin of the **neutrino**, a tiny particle that rarely touches normal matter. Scientists long thought it was too weakly connected to matter to count.

But the researchers found that it breaks down very slowly. So it lived long enough to briefly rule the young universe's energy as the universe cooled after the **Big Bang**. That may have changed the temperature when tiny particles met their opposites, with help from the **Higgs boson**. That temperature worked like a switch, shaping how a little extra matter survived.

Study author Dr. Rishav Roshan said, "The fact that we are here at all proves that an **asymmetry** between matter and **antimatter** occurred." The idea was published in Physical Review Letters. It is a new idea, not proof.

Word Treasure

antimatter -- The opposite of matter; it destroys matter when they meet.
neutrino -- A tiny particle that rarely touches normal matter.
asymmetry -- A difference between two amounts that should have been equal.
Higgs boson -- A tiny particle that helps give other particles mass.
Big Bang -- The hot beginning of the universe.

Background you need

Matter is the stuff everything is made of, and antimatter is its opposite.

When matter and antimatter meet, they destroy each other.

Scientists study tiny particles to learn how the universe began.

Break it down

WHO: Researchers from the University of Southampton and the Indian Institute of Technology Kanpur

WHAT: They have a new idea about why the universe exists

WHERE: In the young universe, just after the Big Bang

WHY: A forgotten particle may have let a little extra matter survive

Why it matters

Asking why anything exists at all is one of the biggest questions you can wonder about.

Test what you remember

1. What was made in almost equal amounts long ago?

- (A) Matter and antimatter
- (B) Stars and planets
- (C) Heat and cold
- (D) Neutrinos and Higgs bosons

2. What happens when matter meets antimatter?

- (A) They live longer
- (B) They make stars
- (C) They destroy each other
- (D) They cool down

3. What is the forgotten particle?

- (A) The Higgs boson
- (B) A neutrino
- (C) Antimatter
- (D) A heavier cousin of the neutrino

4. Why was the particle long thought not to count?

- (A) It was too big
- (B) It broke down too fast
- (C) Too weakly connected to matter
- (D) It was too heavy

5. What did the researchers find about the particle?

- (A) It breaks down very fast
- (B) It is proof
- (C) It breaks down very slowly
- (D) It touches normal matter often

6. Who is the study author quoted in the story?

- (A) Physical Review Letters
- (B) University of Southampton
- (C) Dr. Rishav Roshan
- (D) Higgs boson

Answer key (try the quiz first!): 1: A 2: C 3: D 4: C 5: C 6: C

Questions to think about

1. Dr. Rishav Roshan: Our existence proves an asymmetry between matter and antimatter occurred.

MY THOUGHTS (at least 20 words):